

HEALTHCONNECT EXPERIENCE LAB

WEEK 7 TESTING & VALIDATION PLAN

Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Document Control

Field	Detail
Project	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Programme	AnalystLab Africa Experience Lab
Workstream	Project Management
Testing Phase	Week 7 – Testing, Refinement & End-to-End Validation
Version	1.0
Status	Controlled Working Document

1. PURPOSE

This plan establishes the approach, controls and acceptance framework for testing the HealthConnect solution as it progresses from workstream outputs to integrated validation.

It is aligned with ISTQB test-management principles and incorporates the Week 7 project requirements, available test evidence, approved project resources and identified cross-track dependencies.

The plan is intended to determine whether:

- Individual outputs meet their defined requirements;
- Data and processing remain reliable throughout the pipeline;
- Model behaviour is sufficiently characterised for its intended use;
- Interfaces correctly transfer and consume dependent outputs;
- Identified defects are corrected and successfully retested;
- Residual risks and limitations are sufficiently understood for Week 8 progression.

2. TEST BASIS AND SCOPE

2.1 Test Basis

Testing will be derived from:

- HealthConnect project objectives and intended workflow;
- Week 7 testing requirements;
- Week 6 workstream outputs and identified dependencies;
- HealthConnect_Appointment_Data.csv;
- HealthConnect_Data_Dictionary.xlsx;
- HealthConnect_Clinic_Knowledge_Base.docx;
- Model evaluation and pipeline evidence;
- Previously identified defects, limitations and integration conditions.

Original approved resources remain unchanged; derived test artefacts will be separately identified and traceable to their source.

2.2 Scope

Test Domain	Principal Test Items
Data & Analytics	Data integrity, schema, missingness, duplicates, feature calculations, leakage controls, KPIs, dashboard behaviour and analytical findings
Data Science	Baseline/candidate performance, cross-validation, error analysis, segment behaviour, feature contribution, threshold sensitivity and overfitting
ML Engineering	Input validation, preprocessing, model interface, scoring, negative scenarios, reproducibility, regression and pipeline behaviour
Generative AI	Knowledge grounding, emergency/escalation behaviour, ambiguity, unsupported requests, clinical boundaries, consistency and privacy-sensitive scenarios
Integration	Analytics → DS, DS → ML, ML → decision/intervention workflow and other material interfaces
End-to-End	Integrated flow from data through model output to decision/intervention where the required components are available

Clinical efficacy, production performance and real-world generalisability beyond the available evidence are outside the scope of this validation phase.

3. TEST APPROACH

Testing will follow a risk-based, incremental and evidence-driven approach, progressing from component testing through interface testing to end-to-end validation.

Test Levels

Level	Definition
Component Testing	Validation of individual workstream outputs.
Integration Testing	Validation of interfaces and dependencies between workstreams.
End-to-End Validation	Validation of the integrated HealthConnect workflow once the necessary interfaces and downstream logic are available.

Test Types

Testing will apply the appropriate combination of:

- Functional testing;
- Data-quality and integrity testing;
- Negative and boundary testing;
- Regression testing;
- Model performance and robustness evaluation;
- Interface/compatibility testing;
- Usability and interpretability assessment;
- Exploratory testing where structured investigation may expose unexpected behaviour.

Testing priority will be driven by patient-facing impact, decision consequences, data integrity, integration criticality, downstream dependency and Week 8 readiness.

4. TEST CONDITIONS AND ACCEPTANCE FOCUS

4.1 Data and Analytics

Testing will verify:

- Source-data completeness, duplicates and schema consistency;
- Alignment with the Data Dictionary;
- Correctness of engineered features, including point-in-time history;
- Absence of material information leakage;
- KPI reconciliation against underlying data;
- Dashboard calculations, filters and visual representation;
- Consistency and interpretability of analytical findings.

Analytical observations will remain descriptive/associational unless supported by evidence sufficient for stronger inference.

4.2 Data Science

Testing will assess:

- Baseline versus candidate performance;
- Cross-validation stability;
- Temporal/backtesting behaviour where applicable;
- Precision, recall and ROC-AUC;
- False-positive and false-negative patterns;
- Segment performance;
- Feature ablation/contribution;
- Threshold sensitivity;
- Over-fitting indicators.

Current evidence includes 5-fold stratified CV ROC-AUC of 0.6771 ± 0.0090 , refined-model ROC-AUC of 0.6668, recall of 0.6742, and an independent threshold evaluation of approximately 0.83 recall / 0.56 precision.

These results characterise the candidate but do not constitute operational acceptance. Threshold selection remains subject to downstream decision and intervention requirements.

4.3 ML Engineering

Testing will cover the following chain:



Validation includes expected/invalid inputs, schema and preprocessing compatibility, probability outputs, reproducibility, error handling, regression and batch behaviour.

Current technical evidence includes 69/69 automated tests passing, 948/948 records batch-scored, reproducibility testing, negative-scenario testing and regression following correction of a date-range defect.

The previously integrated Random Forest demonstrates pipeline/interface capability only; it is not evidence of final Data Science candidate integration.

4.4 Generative AI

Scenario testing will cover:

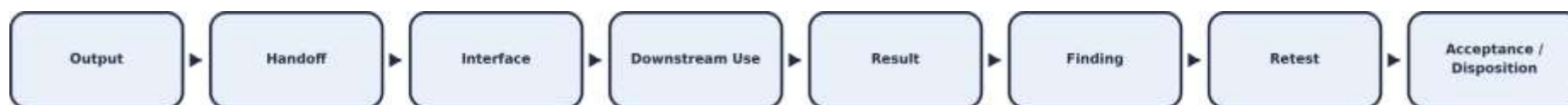
- Normal clinic-support requests;
- Scheduling urgency;

- Genuine and indirect emergency language;
- Unsupported requests;
- Diagnosis and treatment/medication boundaries;
- Knowledge Base grounding;
- Consistency;
- Escalation;
- Privacy-sensitive scenarios;
- Service-availability conditions.

Eight scenarios have been live-tested cumulatively. Two planned Week 7 scenarios were blocked by service availability and remain untested, not failed.

5. INTEGRATION AND INTERFACE VALIDATION

Integration testing will verify the transfer, interpretation and downstream use of material outputs. The governing traceability chain is:



Interface	Validation Requirement	Current Position
Analytics → Data Science	Relevant analytical evidence is received, interpreted and appropriately incorporated or dispositioned	Downstream incorporation requires confirmation
Data Science → ML Engineering	Final candidate, 33-feature schema, preprocessing assumptions and configuration are transferred and tested	Pending
ML → Decision/Intervention	Model output is translated into an agreed decision and intervention pathway	Pending
GenAI → HealthConnect Context	Behaviour remains within approved knowledge and defined governance boundaries	Component evidence available; broader integration pending

Communication or handoff alone does not constitute integration acceptance.

6. TEST ENVIRONMENT, DATA AND TESTWARE

Testing will use controlled project resources and appropriate execution environments.

Approved Resources

- HealthConnect_Appointment_Data.csv
- HealthConnect_Data_Dictionary.xlsx
- HealthConnect_Clinic_Knowledge_Base.docx

Controlled Testware

- Testing Activity Tracker;
- Test scenarios and expected-result criteria;
- Test evidence;
- Integration & Issue Log;
- Risk & Dependency Register;
- Decision Log;
- Model and pipeline evaluation outputs;
- Dashboard and GenAI evidence.

The Testing Activity Tracker is the execution record; this plan establishes the framework against which those activities are performed.

7. ENTRY CRITERIA

A test item may enter execution when the applicable conditions are met:

1. The item is available and sufficiently stable;
2. The relevant test basis is identifiable;
3. Required data, environment and dependencies are available;
4. Expected behaviour or acceptance criteria are defined;
5. Required testware is available;
6. Material constraints are documented.

Where these conditions are not met, the activity will be recorded as Pending, Blocked or Not Testable, as applicable.

8. DEFECT, RETEST AND CHANGE CONTROL

Material findings will be recorded, assessed, assigned and tracked through resolution. Each finding will have:

- An identifiable affected component/interface;
- Severity and impact assessment;
- Owner;
- Corrective/refinement action;
- Retest requirement;
- Evidence of disposition.

A correction does not constitute closure until appropriate retesting demonstrates that the expected behaviour has been restored or the residual risk has been formally accepted.

Changes affecting validated data, features, model configuration, interfaces, prompts, Knowledge Base grounding or workflow logic will receive appropriate regression assessment.

9. TEST MONITORING AND REPORTING

Testing progress will be monitored against:

- Planned versus executed activities;
- Status and coverage of critical tests;
- Unresolved findings;
- Retest completion;
- Interface/dependency status;
- Evidence completeness;
- Material changes and residual risk.

Project Management will consolidate the testing position and escalate conditions affecting critical dependencies, acceptance or Week 8 progression.

10. ROLES AND RESPONSIBILITIES

Role	Responsibility
Project Management	Test coordination, dependency control, evidence traceability, issue/risk escalation, reporting and readiness assessment
Data Analytics	Data-quality, KPI, dashboard and analytical validation
Data Science	Model performance, robustness, error and candidate validation
ML Engineering	Pipeline, interface, scoring, reproducibility and regression testing

Role	Responsibility
Generative AI	Scenario, grounding, boundary, escalation and consistency testing
Cross-Track Owners	Validation of material interfaces and downstream use of dependent outputs

Technical findings remain owned by the relevant technical workstream; Project Management governs coordination, traceability and progression control.

11. EXIT CRITERIA

A component or test scope may be considered complete when:

- Critical planned conditions have been executed;
- Required evidence is available and traceable;
- Material findings are resolved or formally dispositioned;
- Required retesting is complete;
- Known limitations are documented.

Week 7 overall validation will require sufficient evidence of:

1. Critical component testing;
2. Material defect disposition and regression testing;
3. Validation of critical cross-track interfaces;
4. Actual Data Science → ML candidate integration;
5. Assessment of model-output decision/intervention logic;
6. Disposition of material GenAI testing gaps;
7. End-to-end validation to the achievable project scope;
8. Documented residual risks and limitations;
9. Evidence sufficient to support the Week 8 readiness decision.

Unmet criteria remain open or formally carried forward; they are not treated as implicit acceptance.

12. CURRENT VALIDATION POSITION

The available evidence demonstrates substantial component-level testing and refinement across the technical workstreams.

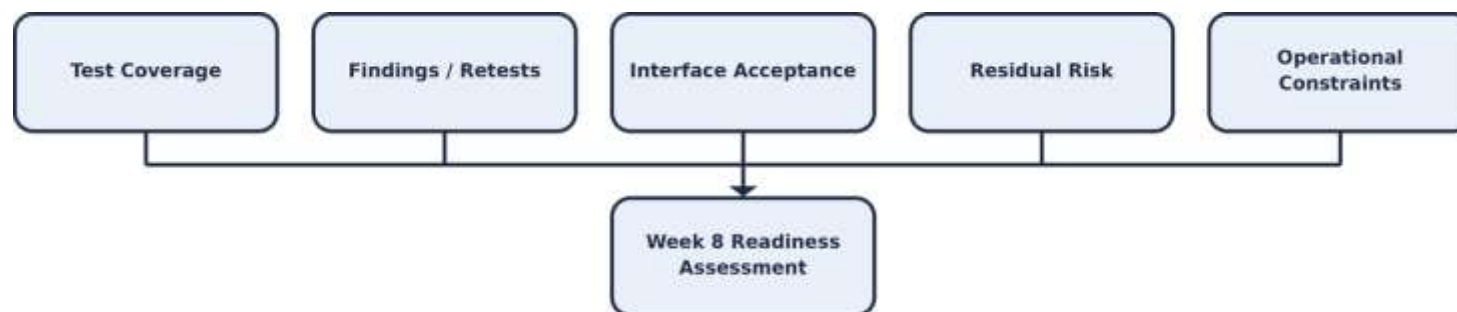
The principal conditions still affecting integrated acceptance are:

- Final Data Science candidate transfer and ML integration;
- Feature-schema/preprocessing compatibility;
- Threshold and decision-rule acceptance;
- Score → decision → intervention validation;
- Analytics downstream disposition;
- Outstanding GenAI scenario, consistency and privacy coverage;
- End-to-end workflow validation;
- Operational readiness and residual-risk assessment.

Accordingly, component-level technical evidence currently supports continued integration activity but does not, by itself, establish full-system validation or Week 8 readiness.

13. WEEK 8 READINESS CONTROL

The Week 8 readiness decision will be based on the consolidated evidence from:



The resulting status will be recorded in the Week 8 Readiness Assessment, supported by the Testing Activity Tracker, Integration & Issue Log, Risk & Dependency Register and Decision Log.